

UNIVERSITY OF AZAD JAMMU & KASHMIR CHELLAH
CAMPUS MUZAFFERABAD



DEPARTMENT OF SOFTWARE ENGINEERING

Databases Systems

(Lab Task # 03)

Submitted By:

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(2024-SE-37)

Submitted To:

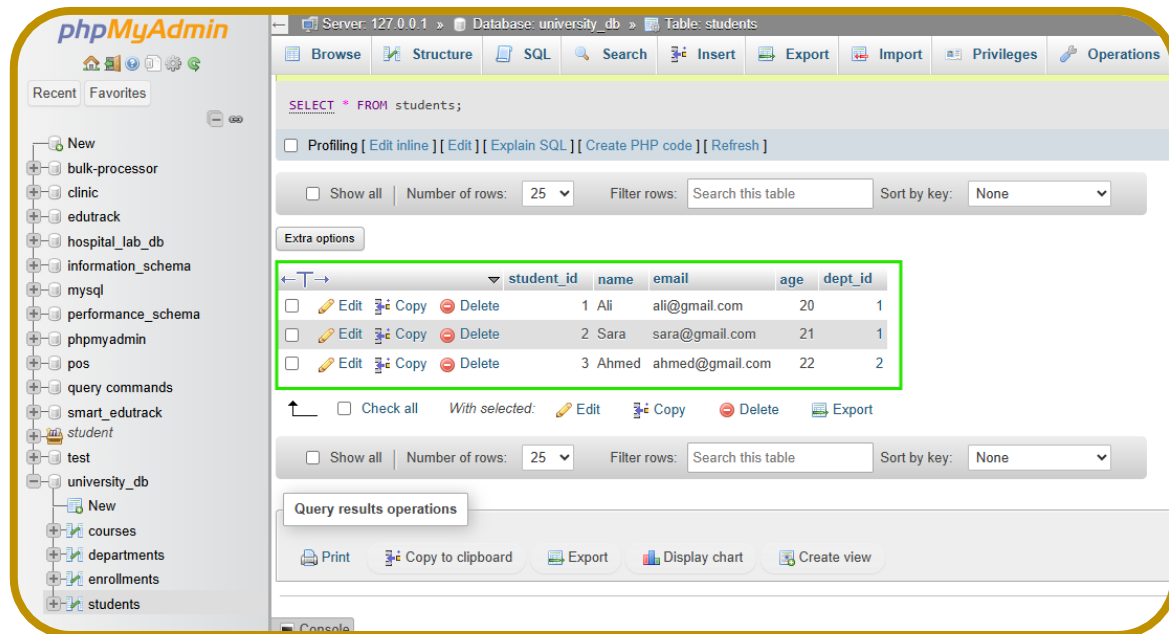
Respected "Engr. Awais Rathore"

Session: 2024-2028

Lab Task: University Database & SQL Keys Practice

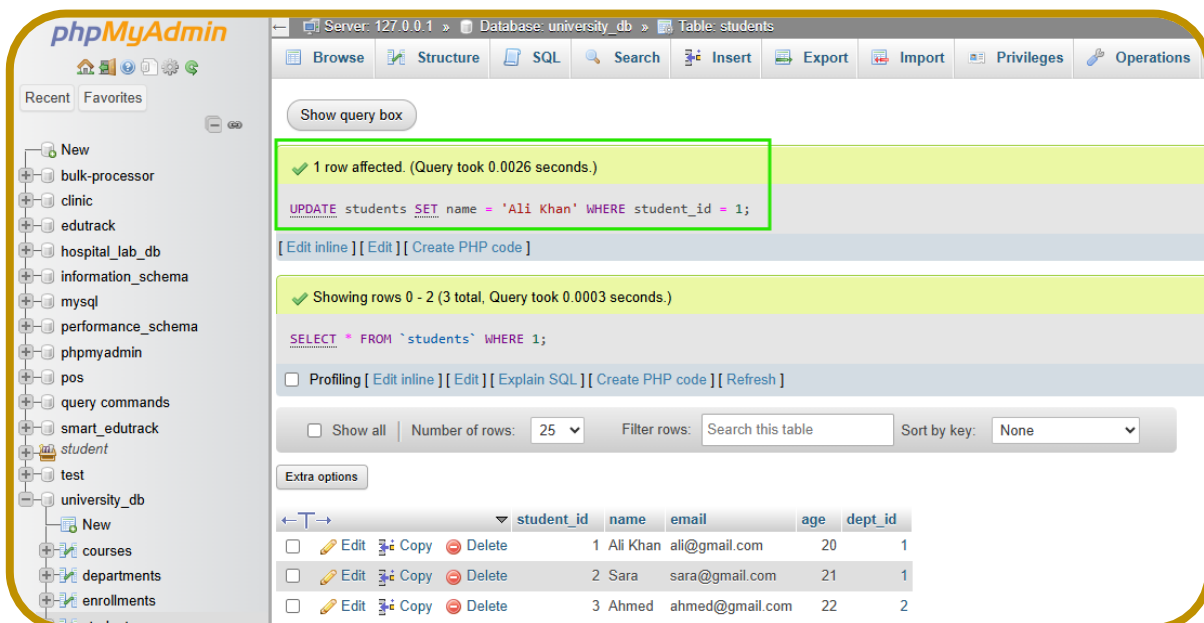
➤ Tasks Performed:

- Data Insertion and Verification



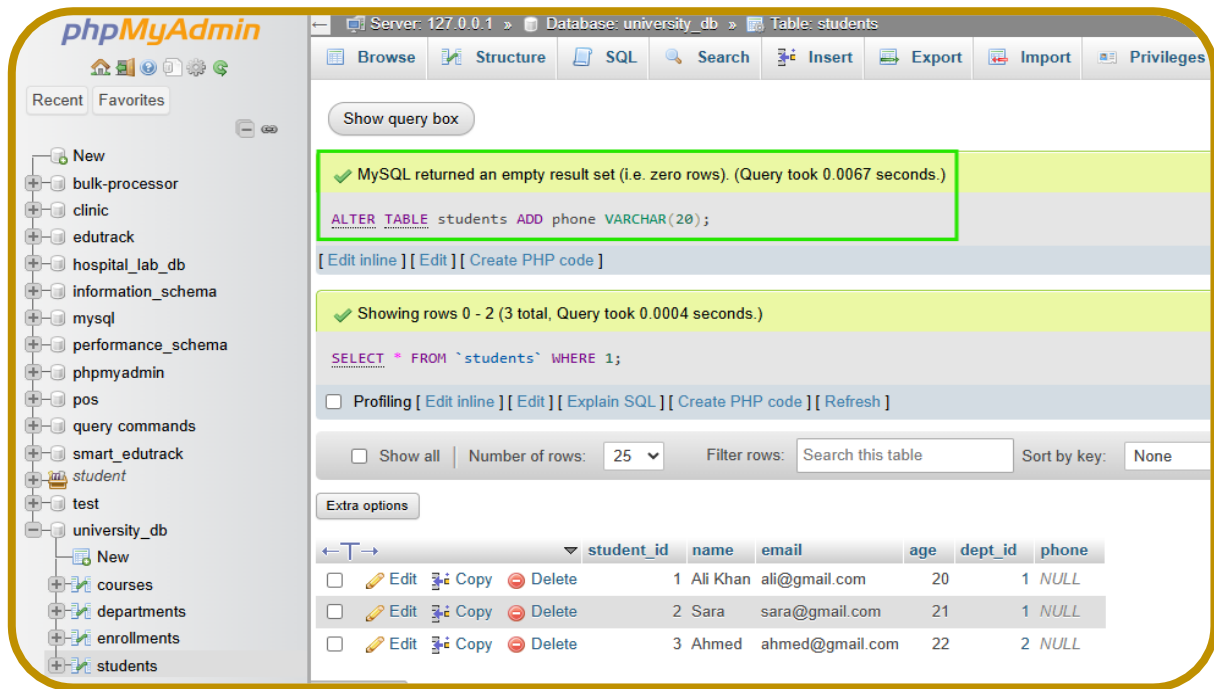
- **Description:** This screenshot verifies that the **INSERT** command was successful. Three records for Ali, Sara, and Ahmed are correctly displayed in the students table.

- Data Manipulation (UPDATE Command)



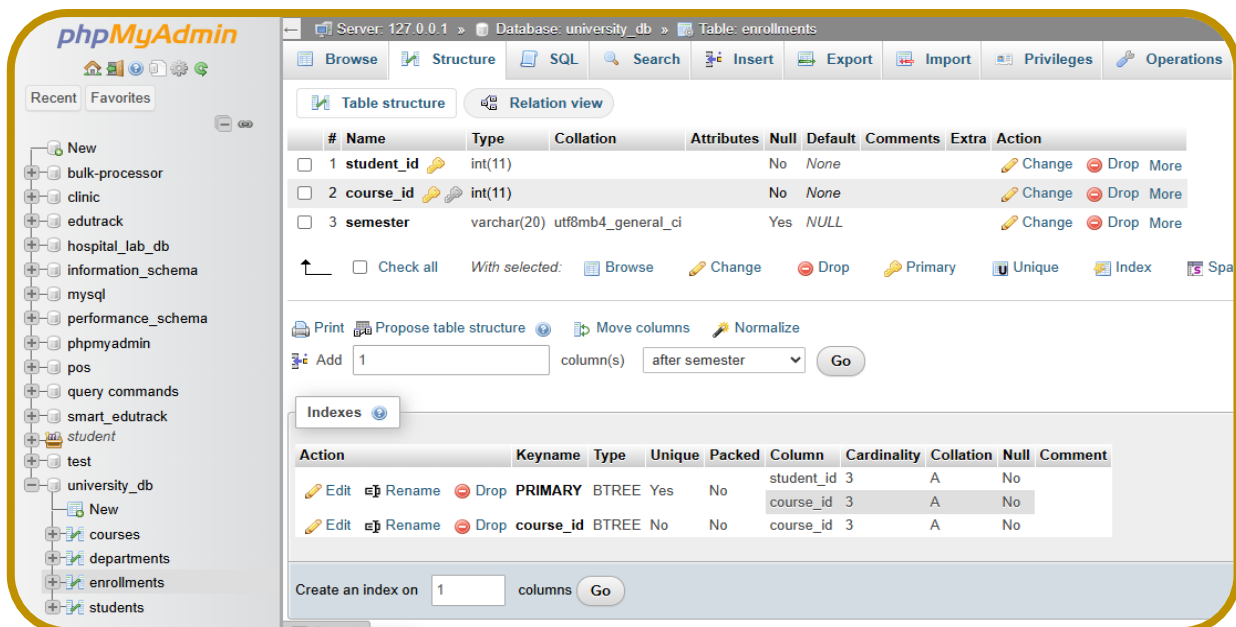
- **Description:** The successful execution is confirmed by the green "1 row affected" notification in phpMyAdmin, and the result is reflected in the updated table.

- Table Modification (ALTER Command)



- Description:** The screenshot shows the successful execution of the **ALTER** query. The updated schema now includes the phone column with default **NULL** values for existing records.

- Composite Key Implementation



- Description:** In the "Structure" tab of the enrollments table, both **student_id** and **course_id** are marked with key icons. The "Indexes" section further confirms that both columns together form the **PRIMARY** index of the table.

➤ Data Removal (DELETE & TRUNCATE Commands)

- **DELETE Command:**

Successfully practiced removing specific records using the WHERE clause (e.g., deleting a student by student_id). This command was used to maintain data accuracy without affecting the table structure.

- **TRUNCATE Command:**

Performed a Truncate operation on a dummy table to understand how it differs from DELETE. It was noted that TRUNCATE quickly removes all rows and resets the auto-increment counter, making it more efficient for clearing entire tables.

- **Observation:**

Verified that while **DELETE** is a DML command (can be rolled back), **TRUNCATE** is a DDL command that permanently resets the table data while keeping the schema intact.